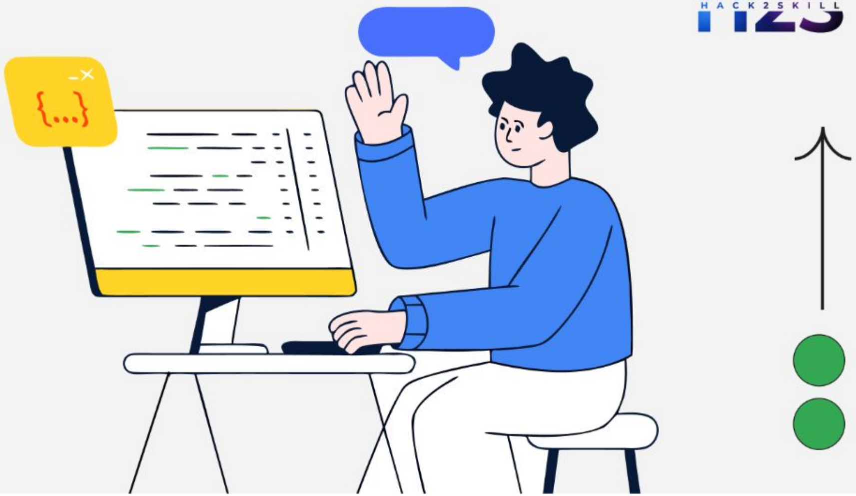
# Solution Challenge



## Team Details

**Team name: DevDrops**
**Team leader name: Aryan Jha**
**Problem Statement: [Rapid Crisis Response] Accelerated Emergency Response and Crisis Coordination in Hospitality**

## **Beacon:** AI- Powered Crisis Response

**Problem:** Crises in large venues fail due to language barriers, misinformation, and slow communication between guests and staff.

**Solution:** Beacon is a unified, real-time emergency ecosystem that bridges the critical gap between a cry for help and the arrival of a first responder. By automating threat analysis and translation, we turn chaotic panic into coordinated action instantly.

**The Magic**: We leverage Google's Gemini AI to instantly translate distress signals, categorize threats, and generate actionable protocols—synchronizing guests, staff, and first responders in under 100 milliseconds.

## Redifining Emergency Infrastructure : Why Legacy Systems fail?

- **The Status Quo**: Current systems rely on fragmented communication—panic buttons that provide zero context, slow 911 phone calls, and siloed radio chatter (walkie-talkies).

- **The Beacon Difference**:
  Legacy: "Room 305 alarmed." (No context, blind response).
  Beacon: "Medical emergency in Room 305. Translated from Spanish: 'Chest pains'.
  Suggested Action: Dispatch AED immediately."

- **Proactive vs. Reactive**: We replace rigid, manual reporting with an AI-enriched, automated flow.

# The Beacon Architecture : An End to End Resolution

1. **Frictionless Ingestion**: Guests use a mobile SOS portal to report emergencies in natural language or via one-tap quick actions.

2. **AI Enrichment (Gemini Engine)**: Unstructured panic data is instantly ingested, translated, and structured into a professional threat assessment and protocol checklist.

3. **Omnichannel Synchronization**:
Command Center gets geographic live-mapping.
Staff get rich-text context on their devices.
Responders get high-contrast dispatch pagers.

# Our Unique Selling Proposition : Intelligence meets Immediacy

- **Sub-100ms Omnichannel Sync**: Custom Socket.io nodes guarantee that critical data displays across all active devices simultaneously without page refreshes.

- **Real-Time Linguistic AI:** Utilizing Gemini 2.5 Flash to eliminate the single deadliest delay in global hospitality: the language barrier.

- **Role-Specific Dashboards**: Instead of one cluttered screen, Beacon delivers specialized UI views tailored exactly to what a Manager, a Staff Member, or a Medic needs to see.

## Comprehensive Feature Suite : An Interface for Every Responder

📱 **Guest SOS Portal:** Frictionless mobile interface with one-tap emergency triggers and natural-language chat.

🧠 **Gemini AI Core Engine:** Autonomous translation, severity scoring, and instant generation of standard operating protocols.

🗺️ **Live Command Center:** Interactive, geographic map (powered by Leaflet) with pinpoint radar locations for security management.

🚑 **Emergency Dispatch Pager:** Full-screen, high-contrast critical alert view designed for medics and armed response teams.

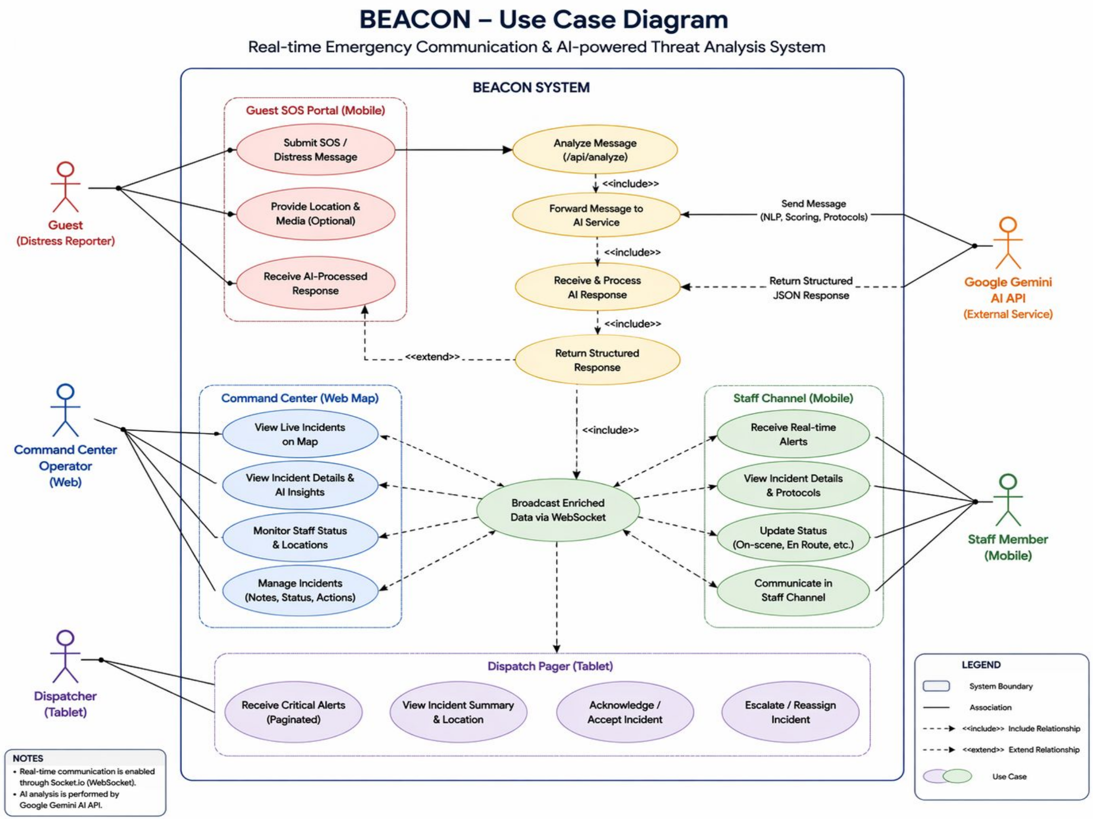
📡 **Staff Alert Channel:** WhatsApp-style group feed broadcasting real-time situational awareness to all ground staff.

Build with AI

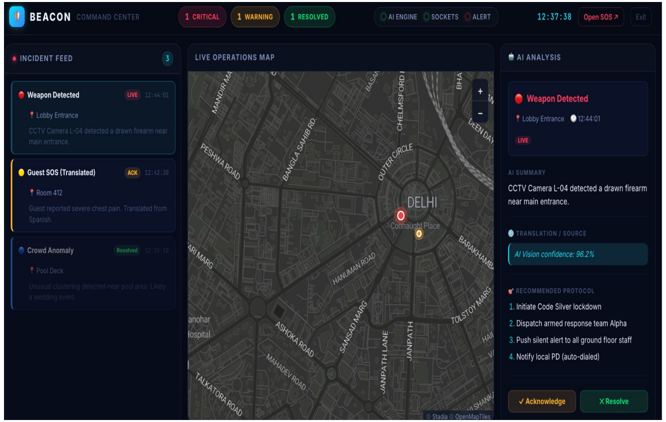
Solution Challenge

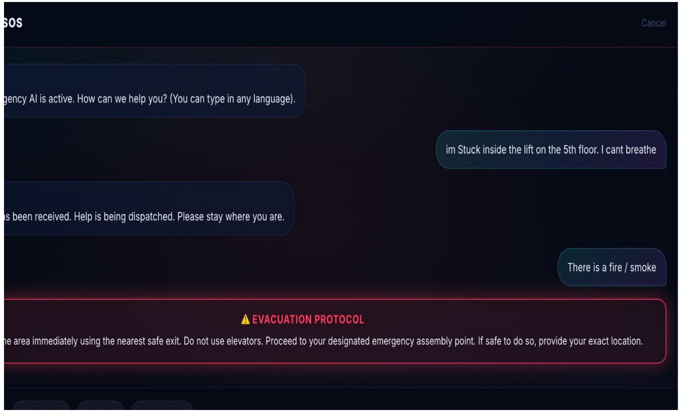
H2S HACK2SKILL

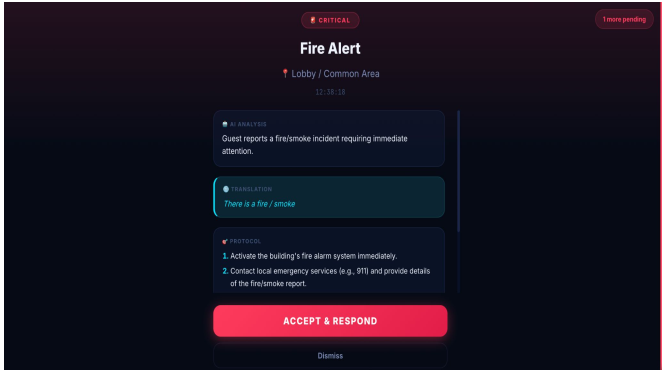
# Use-case diagram: Beacon Platform

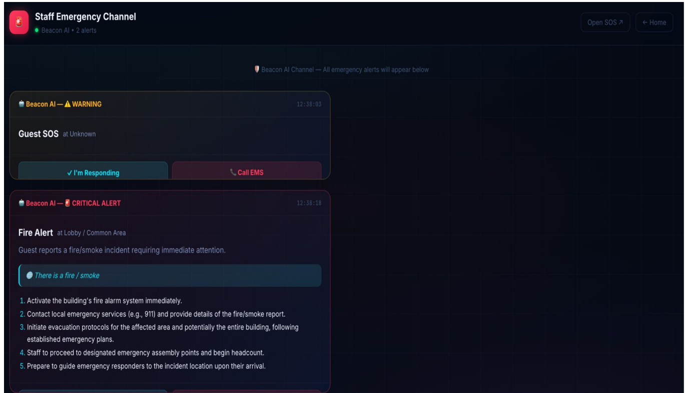


# Specialized Operational Views : High Fidelity UI Dashboard

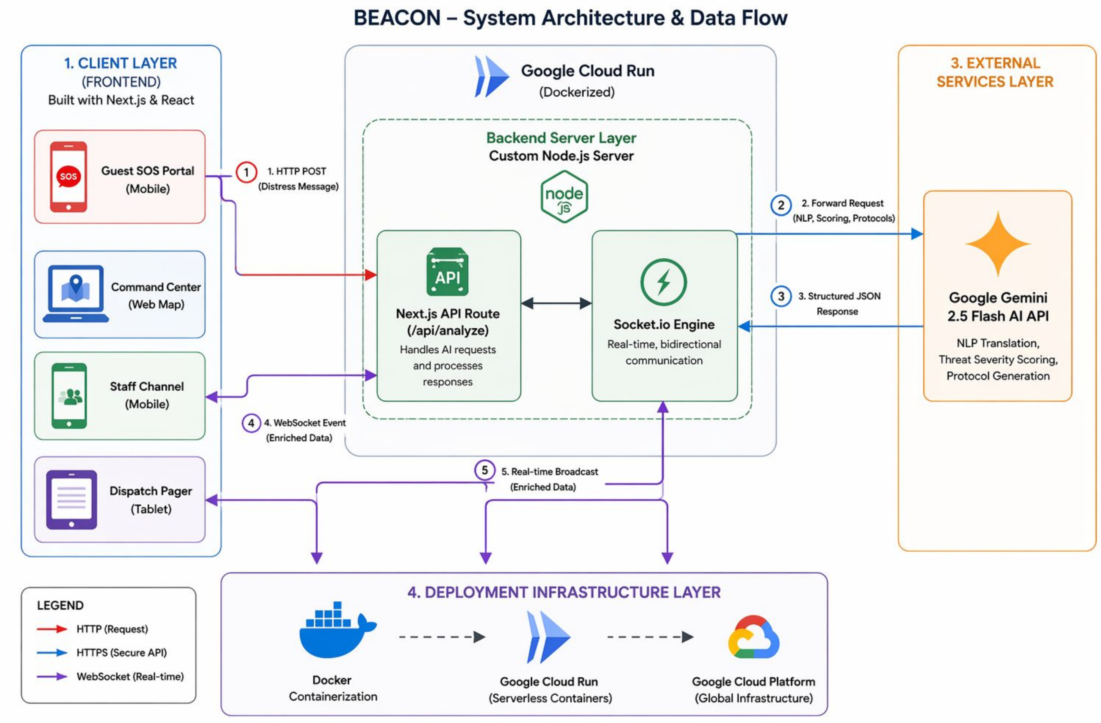


# Architecture diagram



# Our Tech Stack : Built for Speed, Scale and Reliability

- **Frontend & UI**
  Next.js 14 & React: For scalable, component-based view rendering.
  TypeScript: For strict type-safety across the ecosystem.
  CSS Modules: For custom glassmorphic styling without heavy component libraries.

- **Real-time Engine**
  Node.js (Custom Server): To host the API logic and WebSocket layer.
  Socket.io: Facilitating the sub-100 millisecond bidirectional event syncing across all devices.

- **Geographic Data**
  Leaflet & React-Leaflet: Pure open-source interactive mapping.
  Stadia/OpenMapTiles: Providing the dark-mode, high-contrast geographic tile layers.

- **AI & Infrastructure**
  Google Gemini 2.5 Flash: Powering the NLP, translation, and generative protocols.
  Docker: Containerizing the combined Next.js and Node environment.
  Google Cloud Run: Providing auto-scaling, serverless deployment with native WebSocket support.

## Estimated implementation cost (MVP Phase) : Pay as you Go Architecture

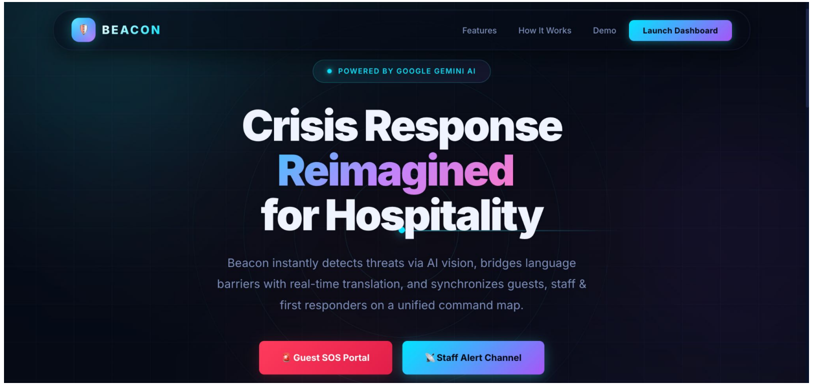
- **Hosting & Compute (Google Cloud Run):**
  Cost: ~$5 to $15 / month (Free tier heavily subsidizes this).
  Why: We use a Dockerized serverless setup. We only pay for the exact milliseconds the server is processing an emergency.

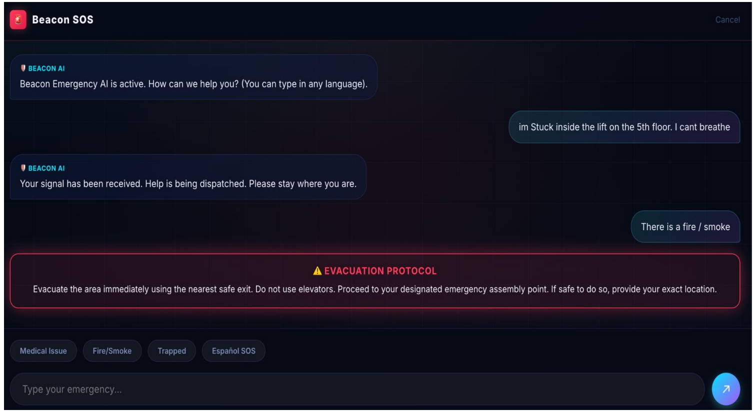
- **AI Processing Engine (Google Gemini 2.5 Flash):**
  Cost: ~$0.07 per 1 Million Tokens (Pennies per month).
  Why: By specifically choosing the "Flash" model over Pro, we get blazing-fast translation and protocol generation at a fraction of the cost.

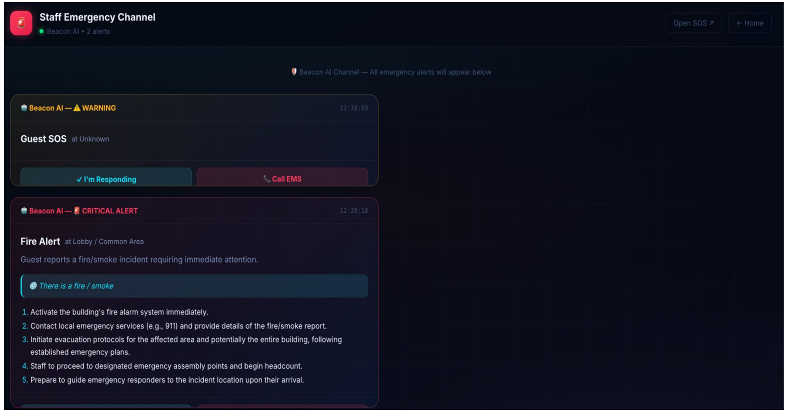
- **Real-Time Architecture (Socket.io):**
  Cost: $0 (Open Source).
  Why: Instead of paying thousands a year for enterprise WebSocket services like Pusher, we self-host our own Node engine inside Cloud Run.

- **Geographic Mapping Layers (Leaflet & OpenMapTiles):**
  Cost: $0 (Open Source / Free Tier up to 2.5M requests).
  Why: We bypassed expensive Google Maps API keys by using open-source mapping engines.

# Snapshots of MVP



**Beyond the MVP : Scaling the Beacon System**

👁️ **Vision AI Integration (CCTV):** Connecting Beacon directly to existing hotel security cameras, allowing Google Gemini 1.5 Pro to visually ingest video feeds and trigger autonomous alerts without human input.

📞 **Automated Emergency Dispatch:** Integrating Voice AI to automatically place phone calls to local 911 dispatchers with the translated SOS data and exact GPS coordinates.

⌚ **Smart Wearable Integration:** Syncing with guest wearables (e.g., Apple Watch, Fitbit) for automatic biometric triggers like heavy fall detection or abnormal heart rates in private rooms.

📊 **Predictive Analytics Engine:** Migrating from an in-memory database to a persistent cloud database to analyze years of incident history, generating heat maps of danger zones to prevent future crises.

# LINKS

**GitHub Public Repository:** https://github.com/Aryans06/Beacon

**Demo Video Link (Youtube):** https://youtu.be/Lq2CQXXIsEo

**MVP Link :** https://beacon-682002074948.us-central1.run.app/

**Working Prototype Link :** https://beacon-682002074948.us-central1.run.app/

Build with AI

H2S

Solution Challenge

# Thank You